Robert M. Gower

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modern optimization **adaptive methods** **stochastic optimization** **machine learning** **randomized numerical linear algebra**

Profile

I work on the mathematics of modern optimization, especially Adam, Muon, Polyak step-sizes, and non-Euclidean gradient descent. My research sits between optimization theory and machine-learning practice, with an emphasis on simple algorithms that exploit structure.

Appointments and Education

2021–present	**Research Scientist** , Flatiron Institute Center for Computational Mathematics, New York
2017–2021	**Assistant Professor** , Telecom Paris, Institut Polytechnique de Paris Machine learning and optimization
2021	**Visiting Scientist** , Google Research New York
2020	**Visiting Scientist** , Facebook Research New York
2016–2017	**Laureate Fellow** , Fondation Sciences Mathematiques de Paris ENS and Inria, Paris
2016	**Ph.D.** , University of Edinburgh Optimization and Operations Research
2011	**M.Sc.** , University of Campinas Applied Mathematics

Recent Highlights

2026	*The Polar Express* received the ICLR Outstanding Paper Honorable Mention and was presented as an oral.
2026	Upcoming talk at the SIAM Conference on Optimization (OP26), June 2–5.
2025	*In Search of Adam's Secret Sauce* was an oral presentation at NeurIPS.

Selected Papers

- N. Amsel, D. Persson, C. Musco, **R. M. Gower**. *The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm*. ICLR 2026, oral; Outstanding Paper Honorable Mention.
- A. Orvieto, **R. M. Gower**. *In Search of Adam's Secret Sauce*. NeurIPS 2025, oral.
- F. Schaipp, R. Ohana, M. Eickenberg, A. Defazio, **R. M. Gower**. *MoMo: Momentum Models for Adaptive Learning Rates*. ICML 2024.
- **R. M. Gower**, D. Goldfarb, P. Richtarik. *Stochastic Block BFGS: Squeezing More Curvature out of Data*. ICML 2016.
- **R. M. Gower**, P. Richtarik. *Randomized Iterative Methods for Linear Systems*. SIAM J. Matrix Anal. Appl., 2015; Leslie Fox Prize, 2nd place.

Honors, Service, and Mentoring

- ICLR Outstanding Paper Honorable Mention, 2026; Leslie Fox Prize, 2nd place, 2017; FSMP Laureate Fellowship, 2016; reviewer awards at ICML and NeurIPS.
- Area chair for NeurIPS and TMLR; reviewer for ICML, JMLR, SISC, SIOPT, Mathematical Programming, and related venues.
- Mentored Ph.D. students and interns including Nidham Gazagnadou, Rui Yuan, Samy Jelassi, Si Yi Meng, Fabian Schaipp, Slavomir Hanzely, Aaron Mishkin, David Persson, Michael Crawshaw, and Tetiana Parshakova.